

Priya Nair

priyanair.eng@gmail.com | +91 9012345678 | LinkedIn | GitHub

Summary

Electronics and Computer Science undergraduate with strong embedded systems and IoT development background. Built 4+ hardware-software integrated projects; proficient in ARM Cortex programming and RTOS. Looking to bridge the gap between hardware and intelligent software.

Education

B.Tech in Electronics & Computer Science 2022–2026

NIT Calicut **CGPA: 8.9**

Class XII (2022): 97% | Class X (2020): 99%

Technical Skills

Languages: C, C++, Python, MATLAB, Verilog

Embedded: ARM Cortex-M, Arduino, ESP32, Raspberry Pi, STM32, FreeRTOS

IoT & Protocols: MQTT, I2C, SPI, UART, BLE, LoRa

Tools: Keil MDK, PlatformIO, Altium Designer, Git, MATLAB Simulink

ML on Edge: TensorFlow Lite, Edge Impulse

Projects

Smart Agriculture IoT System ESP32, MQTT, Python, AWS IoT

- Deployed sensor network monitoring soil moisture, temperature, and humidity across 5 nodes
- Integrated MQTT broker with AWS IoT Core for real-time cloud data ingestion
- Reduced water usage by estimated 30% in pilot greenhouse setup

Edge AI Fall Detection Wearable Arduino Nano 33 BLE, TensorFlow Lite

- Trained gesture classification model achieving 94% accuracy on-device
- Deployed model using TensorFlow Lite on Cortex-M4; power consumption under 50mW

RTOS-Based Drone Flight Controller STM32, FreeRTOS, C

- Implemented PID control loops for quadrotor stabilization using FreeRTOS tasks
 - Achieved stable hover within $\pm 2^\circ$ deviation in bench tests
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Experience

Embedded Systems Intern — Bosch India Ltd. May 2025 – Aug 2025

- Contributed to firmware development for automotive sensor modules in C
 - Wrote unit tests and participated in hardware-in-the-loop simulation testing
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Certifications

- Embedded Systems — Coursera (University of Colorado)
- IoT Specialization — NPTEL
- MATLAB Onramp — MathWorks